

Visuospatial Perspective Taking in Multimodal Language Models

Jonathan Prunty^{1,1}, Seraphina Zhang^{1,2}, Patrick Quinn¹, Jianxun Lian³, Xing Xie³ & Lucy Cheke^{1,2}

¹Leverhulme Centre for the Future of Intelligence, University of Cambridge

²Department of Psychology, University of Cambridge

³Microsoft Research Asia

Abstract

As multimodal language models (MLMs) are increasingly used in social and collaborative settings, it is crucial to evaluate their perspective-taking abilities. Existing benchmarks largely rely on text-based vignettes or static scene understanding, leaving visuospatial perspective-taking (VPT) underexplored. We procedurally generate large stimulus batteries for two evaluation tasks adapted from human studies: the Rotating Figure Task, probing perspective-taking across angular disparities, and the Director Task, assessing VPT in a referential communication paradigm. Across both tasks, MLMs show pronounced deficits in Level 2 VPT, with failure patterns indicating reliance on simple mirroring heuristics rather than genuine perspective transformations. These results expose critical limitations in current MLMs' ability to represent and reason about alternative perspectives, with implications for use in collaborative contexts.

Keywords: Artificial Intelligence; Social Cognition; Theory of Mind; Spatial Cognition; Language Understanding

Task Details

This section provides a full description of the two tasks and the interface through which the models interacted with them. Both tasks were implemented in the Inspect AI framework (AI Security Institute, 2024). We used the framework's basic agent, which places the model in a ReAct-style loop: at each step the model may reason about the current stimulus, issue a single tool call, and observe the result before proceeding. The basic agent supplies the following generic system prompt:

You are a helpful assistant attempting to submit the correct answer. You have several functions available to help with finding the answer. Each message may perform one function call. You will see the result of the function right after sending the message. If you need to perform multiple actions, you can always send more messages with subsequent function calls. Do some reasoning before your actions, describing what function calls you are going to use and how they fit into your plan. When you have completed the task and have an answer, call the submit() function to report it.

Tools

Every model was provided with two tools across both tasks. The submit() tool reports the final answer for a sample:

```
{
  "name": "submit",
  "description": "Submit an answer for
    ↪ evaluation.",
  "parameters": {
    "type": "object",
    "properties": {
      "answer": {
        "type": "string",
        "description": "Submitted answer"
      }
    }
  },
  "required": ["answer"],
  "additionalProperties": false
}
```

The think() tool encourages explicit reasoning in models that do not reason spontaneously and additionally aided development and debugging:

```
{
  "name": "think",
  "description": "Use this tool to stop, think
    ↪ and reason about the task at hand.
    ↪ Particularly useful to build plans
    ↪ and reflect on them prior to
    ↪ executing actions.",
  "parameters": {
    "type": "object",
    "properties": {
      "thoughts": {
        "type": "string",
        "description": "The thoughts or
          ↪ reasoning process of the agent."
      }
    }
  },
  "required": ["thoughts"],
  "additionalProperties": false
}
```

¹Corresponding Author: jep84@cam.ac.uk

Rotating Figure Task

For the Rotating Figure Task (RFT), each model received an additional task-specific system prompt describing the scenario and the form of the questions:

You are participating in a visual perspective-taking experiment. You will be shown images containing a person (or arrow) and possibly a number or symbol on the ground nearby. Your task is to answer questions about what you can see in the image, considering both your own perspective as the viewer and the perspective of the person/figure in the image. Pay careful attention to: the orientation and position of figures in the image; whether numbers appear as “6” or “9” depending on viewing angle; spatial relationships (left/right, front/behind) from different perspectives; and what is visible versus hidden from different viewpoints. Answer exactly as requested in the prompt.

Dataset generation The RFT datasets were produced by a Python script that outputs a JSON file containing the full specification of each sample together with its stimulus image. Generation is controlled by a set of parameters. We created two datasets. The first covers Control 1, Control 2, Test 1, and Test 2, and was generated with the parameters below:

```
{
  dataset_size : 3000,
  stimulus_sets : "control_1,control_2,
    ↳ level_1,level_2",
  trial_num : 1000,
  control_proportion : 1,
  use_number : True,
  use_arrow : True,
  image_size : [400, 400],
  figure_scale : 1.5,
  margin : 200,
  fov_angle : 60,
  view_distance : 5000,
  placement_locations : "left,right,front,
    ↳ behind",
  rotation_min : 0,
  rotation_max : 360,
  jitter_range : 20,
  resource_path : "resources",
  background_image : "background_with_colours
    ↳ .jpg",
  seed : 42,
}
```

The second dataset supplied the Test 3 samples. It used the same parameters with three exceptions: the stimulus

sets additionally included level_3; the placement locations additionally included front_left and front_right; and control_proportion was set to 1/3. Both datasets are included in the project repository within a password-protected .zip archive.

Example transcript A representative transcript from the Rotating Figure Task is shown below.

System

You are participating in a visual perspective-taking experiment. You will be shown images containing a person (or arrow) and possibly a number or symbol on the ground nearby.

Your task is to answer questions about what you can see in the image, considering both your own perspective as the viewer and the perspective of the person/figure in the image.

Pay careful attention to:

- The orientation and position of figures in the image
- Whether numbers appear as “6” or “9” depending on viewing angle
- Spatial relationships (left/right, front/behind) from different perspectives
- What is visible vs. hidden from different viewpoints

Answer exactly as requested in the prompt.

System

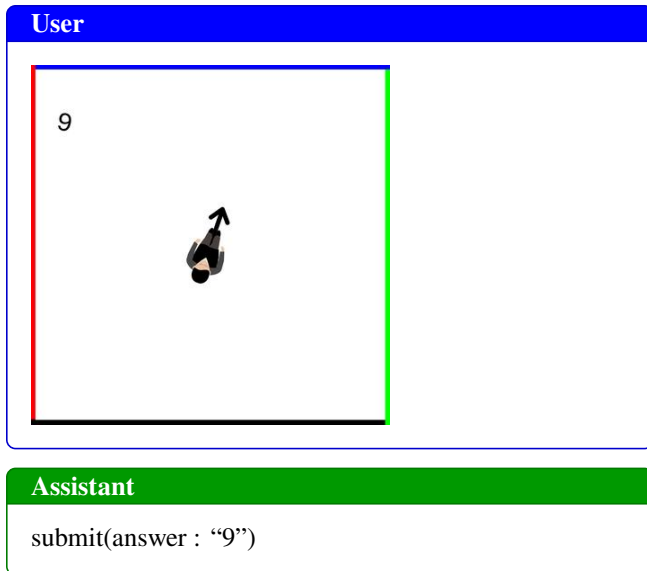
You are a helpful assistant attempting to submit the correct answer. You have several functions available to help with finding the answer. Each message may perform one function call. You will see the result of the function right after sending the message. If you need to perform multiple actions, you can always send more messages with subsequent function calls. Do some reasoning before your actions, describing what function calls you are going to use and how they fit into your plan.

When you have completed the task and have an answer, call the submit() function to report it.

User

There is a person standing in the centre of the image, and a number or letter on the floor nearby. What number or letter can you see in the image?

Respond with a single number or letter:



Director Task

For the Director Task, each model received an additional task-specific system prompt describing the scenario and the response format:

The image displays the front of a shelving grid, which has items stored in grid cells. Your task is to select the correct grid cell that contains the item specified by the user. Each grid cell has a backing plate behind the item that is either transparent or made from an opaque grey material. The user is standing on the opposite side of the grid, viewing it from the back. The user has never seen the grid before, and has no prior knowledge as to the location of the items. Submit your answer in the form <column><row>, for example 'C2', where A is the leftmost column and 1 is the topmost row. Give the coordinates from your point of view.

The model then received the question as a user message, followed by a second user message containing the stimulus image for the sample.

Dataset generation As with the RFT, the Director Task datasets were produced by a Python script that outputs a JSON file containing the full specification of each sample together with its stimulus image. Generation is controlled by the following parameters:

```
{
  grid_width : 4,
  grid_height : 4,
  dataset_size : 1000,
  control_proportion : 0.5,
  items_file : "director_task/items.json",
```

```
  size_prop : 0.25,
  spatial_same_prop : 0.25,
  spatial_diff_prop : 0.25,
  physics_prop : 0.5,
  test_related_item_prop : <related item
    ↪ proportion>,
  control_related_item_prop : <related item
    ↪ proportion>,
  related_blocked_prop : 0.5,
  item_fill_prop : 0.9,
}
```

The related item proportion controls the number of items that match the director's description but must be ruled out on the basis of perspective, and the item fill proportion controls the fraction of grid cells that are occupied. The related item proportion was varied across the values [0.3, 0.5, 0.7, 0.9]. The items file specifies the properties of each item; for example, a book may be given the name `book`, a size of 2, and the properties `stackable`, `red`, and `expensive`. This file, together with the item images, is included in the project repository. The datasets used in our experiments are provided in the repository within a password-protected .zip archive.

Example transcript A representative transcript from the Director Task is shown below.

System

The image displays the front of a shelving grid, which has items stored in grid cells. Your task is to select the correct grid cell that contains the item specified by the user. Each grid cell has a backing plate behind the item that is either transparent or made from an opaque grey material. The user is standing on the opposite side of the grid, viewing it from the back. The user has never seen the grid before, and has no prior knowledge as to the location of the items. Submit your answer in the form <column><row>, for example 'C2' where A is the leftmost column and 1 is the topmost row. Give the coordinates from your point of view.

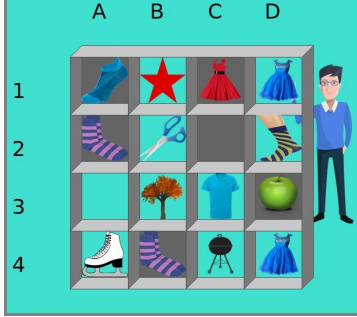
System

You are a helpful assistant attempting to submit the correct answer. You have several functions available to help with finding the answer. Each message may perform one function call. You will see the result of the function right after sending the message. If you need to perform multiple actions, you can always send more messages with subsequent function calls. Do some reasoning before your actions, describing what function calls you are going to use and how they fit into your plan. When you have completed the task and have an answer, call the `submit()` function to report it.

User

Please select the largest star from your point of view

User



Assistant

The largest identified star is located at B1.

Assistant

submit(answer : "B1")

Text-based version To assess how much of the models' error was attributable to the visual processing demands of the task (e.g. correctly identifying the row and column of a given item), we created a text-based version of the Director Task in which the model received a textual rendering of the grid in place of the stimulus image. The same datasets were used for both versions, and all results reported in the main text are for the visual version unless stated otherwise.

In the text-based version, the grid is presented as a Markdown table. The raw representation received by the model is as follows:

```
| Col 0 | Col 1 | Col 2 | Col 3 |
|---|---|---|---|
| A1<br>backing plate:
  ↳ Steel<br>cookie<br>Bool: brown,
  ↳ is_food, is_sweet<br>Scalar: size=1,
  ↳ offset_x=0, offset_y=0 | B1<br>backing
  ↳ plate: Transparent
  ↳ plastic<br>blue_book_small<br>Bool:
  ↳ blue, stackable, book<br>Scalar: size=0,
  ↳ offset_x=0, offset_y=0 | C1<br>backing
  ↳ plate:
  ↳ Glass<br>blue_book_small<br>Bool:
  ↳ blue, stackable, book<br>Scalar: size=0,
  ↳ offset_x=0, offset_y=0 | D1<br>backing
  ↳ plate:
  ↳ Glass<br>blue_book_small<br>Bool:
  ↳ blue, stackable, book<br>Scalar: size=0,
  ↳ offset_x=0, offset_y=0 |
```

```
| A2<br>backing plate:
  ↳ cardboard<br>blue_book_small<br>Bool:
  ↳ blue, stackable, book<br>Scalar: size=0,
  ↳ offset_x=0, offset_y=0 | B2<br>backing
  ↳ plate: Green
  ↳ plastic<br>blue_book_small<br>Bool:
  ↳ blue, stackable, book<br>Scalar: size=0,
  ↳ offset_x=0, offset_y=0 | C2<br>backing
  ↳ plate: Transparent
  ↳ plastic<br>blue_book_small<br>Bool:
  ↳ blue, stackable, book<br>Scalar: size=0,
  ↳ offset_x=0, offset_y=0 | D2<br>backing
  ↳ plate:
  ↳ None<br>blue_book_small<br>Bool: blue,
  ↳ stackable, book<br>Scalar: size=0,
  ↳ offset_x=0, offset_y=0 |
| A3<br>backing plate:
  ↳ None<br>beatbox<br>Bool:
  ↳ Music_instrument<br>Scalar: size=1,
  ↳ offset_x=0, offset_y=0 | B3<br>backing
  ↳ plate: None<br>kitchen_knife<br>Bool:
  ↳ sharp, used_for_cooking<br>Scalar: size=1,
  ↳ offset_x=0, offset_y=0 | C3<br>backing
  ↳ plate: Glass<br>blue_book<br>Bool:
  ↳ blue, stackable, book<br>Scalar: size=1,
  ↳ offset_x=0, offset_y=0 | D3<br>backing
  ↳ plate:
  ↳ None<br>blue_book_small<br>Bool: blue,
  ↳ stackable, book<br>Scalar: size=0,
  ↳ offset_x=0, offset_y=0 |
| A4<br>backing plate: Green
  ↳ plastic<br>Empty_ | B4<br>backing
  ↳ plate: Black
  ↳ plastic<br>black_pot_small<br>Bool:
  ↳ black, hot, used_for_cooking,
  ↳ holds_water<br>Scalar: size=0, offset_x=0,
  ↳ offset_y=0 | C4<br>backing plate:
  ↳ Black
  ↳ plastic<br>blue_book_small<br>Bool:
  ↳ blue, stackable, book<br>Scalar: size=0,
  ↳ offset_x=0, offset_y=0 | D4<br>backing
  ↳ plate: Glass<br>Empty_ |
```

For clarity, the same grid rendered into a more human-readable format is shown in Figure 1.

Model Details

We evaluated four OpenAI models. For each we used the default generation parameters provided by the API:

```
{
  "model": <model_name>,
  "temperature": 1.0,
  "top_p": 1.0,
  "frequency_penalty": 0.0,
  "presence_penalty": 0.0,
  "reasoning_effort": "medium",
```

A1	B1	C1	D1
[BLOCKED]	blue_book_small	blue_book_small	blue_book_small
cookie	B: blue,stackable,book	B: blue,stackable,book	B: blue,stackable,book
S: size:1	S: size:0	S: size:0	S: size:0
A2	B2	C2	D2
[BLOCKED]	[BLOCKED]	blue_book_small	blue_book_small
blue_book_small	blue_book_small		
A3	B3	C3	D3
beatbox	kitchen_knife	blue_book	blue_book_small
S: size:1	S: size:1	S: size:1	S: size:0
A4	B4	C4	D4
[BLOCKED]	[BLOCKED]	[BLOCKED]	Empty
Empty	black_pot_small	blue_book_small	

Figure 1: Human-readable rendering of the text-based Director Task grid shown in the Markdown listing above. Cells marked [BLOCKED] are occluded from the director’s view.

```

}

The reasoning_effort parameter applies only to the rea-
soning models o3 and o4-mini. Model behaviour can change
across a deployment; the specific checkpoints used were:
  • o3-2025-04-16
  • o4-mini-2025-04-16
  • gpt-4o-2024-11-20
  • gpt-4o-mini-2024-07-18

```

Supplementary analyses

Rotating Figure Task

Controls 1 and 2 Before evaluating performance on the test conditions of the *Rotating Figure Task*, we verified that MLMs exhibited the baseline capabilities required by the task. Specifically, Control 1 assessed symbol identity recognition and spatial localisation (e.g., 6/9, Left/Right), while Control 2 assessed basic line-of-sight judgements (e.g., figure is facing the Left/Red wall). Table 1 and Figure 2 summarise the mean accuracy scores across both control tasks, divided by question type (visual and spatial).

Table 1: Mean accuracy in control conditions

Model	Control 1				Control 2			
	Accuracy		Invalid		Accuracy		Accuracy (CR)	
	Visual	Spatial	Visual	Spatial	Visual	Spatial	Visual	Spatial
o3	0.824	0.853	0.048	0.106	0.716	0.752	0.801	0.843
o4-mini	0.886	0.978	0.037	0.002	0.698	0.715	0.774	0.798
gpt-4o	0.912	1.000	0.048	0.000	0.594	0.696	0.663	0.787
gpt-4o-mini	0.901	0.979	0.041	0.000	0.403	0.608	0.435	0.679

CR = corners removed (trials within 10° of a corner [21.8%] excluded)

Invalid = proportion of responses not matching any valid answer option

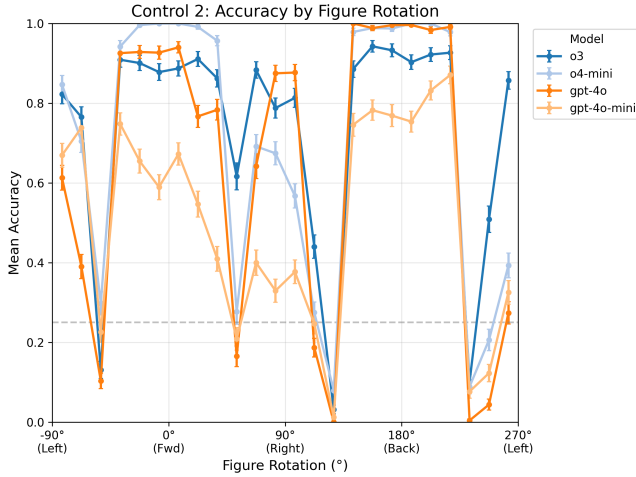


Figure 2: Mean MLM accuracy in Control 2 of the *Figure Rotation Task*. Accuracy is aggregated into 24 bins by figure rotation angle. At 0°, the figure faces forward (shared perspective); at 180°, it faces backward (opposite perspective). Angles wrap around such that 90° corresponds to the figure facing right and -90° (=270°) corresponds to the figure facing left. The dashed line represents chance performance.

MLMs showed high accuracy on Control 1 overall, with gpt-4o, for example, achieving perfect performance on spatial questions. Performance on spatial questions was slightly lower for o3 relative to other models ($M = 0.85$; all other MLMs $M > 0.97$). This difference is partly attributable to o3’s elevated invalid response rate ($M = 0.11$; all other MLMs $M < 0.01$), with o3 frequently producing verbose or irrelevant outputs

despite prompts that explicitly constrained responses (e.g., Left or Right).

Despite the inclusion of an arrow indicating viewing direction in the stimuli, MLMs showed difficulty in reliably parsing the figure’s viewing direction in Control 2. Because this ability was evaluated across the full range of rotation angles, trials in which the figure faced toward image corners (−45°, 45°, 135°, 225°; Figure 2) introduced inherent ambiguity regarding which side of the image the figure was facing. Consistent with this, performance exhibited pronounced decreases at these rotation angles. When ambiguous trials within 10° of a corner (21.8% of trials) were excluded, mean accuracy increased substantially for all models except gpt-4o-mini. This pattern suggests that, unlike the other models, gpt-4o-mini may have more fundamental limitations in extracting viewing direction from static images.

Test 1: Level 1 VPT For the test conditions of the *Rotating Figure Task*, MLM performance was analysed using mixed-effects logistic regression models. Type III Wald χ^2 tests were used to assess main and interaction effects, followed by post-hoc comparisons of estimated marginal means to examine simple main effects across Angular Disparity bins. Model summaries are reported in Tables 5, 6, and 7 for Test conditions 1, 2, and 3, respectively.

Angular Disparity served as the critical manipulation of perspective. It was defined as the absolute angular difference between the participant’s viewing perspective and the figure’s facing direction, constrained to a 0–180° range. Rotations exceeding 180° were subtracted from 360° to yield a symmetric measure of deviation from the forward-facing orientation (0° = facing toward the viewer; 180° = facing away), irrespective of rotational direction. For modelling, Angular Disparity was binned into four 45° intervals (0–45°, 45–90°, 90–135°, and 135–180°), represented by their midpoint values (22.5°, 67.5°, 112.5°, and 157.5°).

In Test 1, symbols appeared either in front of or behind the figure. Visual questions asked whether the figure could see the symbol, whereas spatial questions asked whether the symbol was in front of or behind the figure. Figure 3 shows model accuracy as a function of figure rotation for both question types. Table 2 reports estimated marginal means for Angular Disparity bins and the corresponding simple main effects tests.

Table 2: Mean accuracy in Test 1 by Angular Disparity

Question	Model	0–45°	45–90°	90–135°	135–180°	χ^2	df	p
Visual	GPT-4o-mini	0.386	0.616	0.708	0.772	203.71	3	< .001
	GPT-4o	0.724	0.697	0.766	0.807	22.99	3	< .001
	o4-mini	0.989	0.843	0.860	0.984	145.47	3	< .001
	o3	0.799	0.824	0.803	0.779	4.19	3	0.242
Spatial	GPT-4o-mini	0.580	0.556	0.519	0.528	5.30	3	0.151
	GPT-4o	0.816	0.749	0.724	0.777	16.99	3	< .001
	o4-mini	0.993	0.854	0.866	0.979	128.88	3	< .001
	o3	0.764	0.786	0.782	0.770	1.19	3	0.756

Values are estimated marginal probabilities of correct responses.

Simple main effects test angular disparity within each Model × Question Type combination.

We observed significant main effects of Model, Question Type, and Angular Disparity, as well as significant interactions

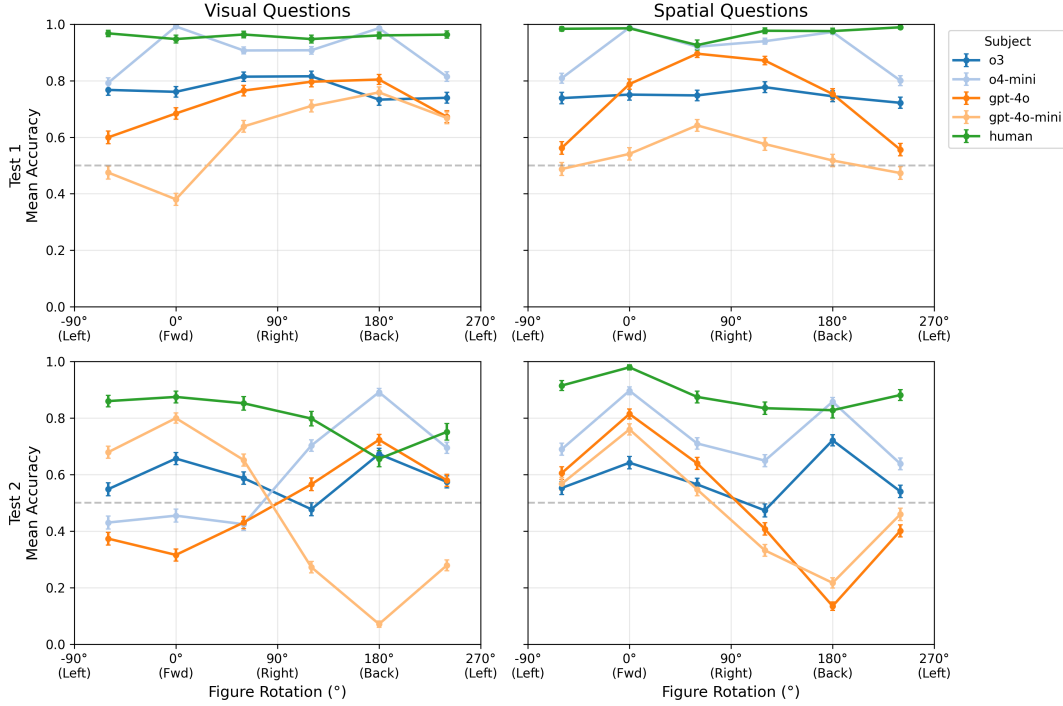


Figure 3: Mean MLM accuracy in Test 1 and 2 of the *Figure Rotation Task*, by Question Type (Visual vs. Spatial). Accuracy is aggregated into 12 bins by figure rotation angle. At 0°, the figure faces forward (shared perspective); at 180°, it faces backward (opposite perspective). Angles wrap around such that 90° corresponds to the figure facing right and -90° (=270°) corresponds to the figure facing left. The dashed line represents chance performance.

among these factors (all $p < .001$; Table 5), indicating that the effect of Angular Disparity differed across models and question types. For GPT-4o-mini, visual question accuracy increased monotonically with Angular Disparity, rising from below chance when perspectives were aligned (0–45°: $M = .39$) to substantially higher accuracy when perspectives were inverted (135–180°: $M = .77$; $\chi^2(3) = 203.71$, $p < .001$). In contrast, spatial question performance for GPT-4o-mini remained near chance across all disparity bins (range $M = .52$ –.58), with no reliable effect of Angular Disparity ($\chi^2(3) = 5.30$, $p = .151$).

GPT-4o and o4-mini both exhibited U-shaped performance profiles, characterised by reduced accuracy at intermediate disparities (45–135°). For GPT-4o, visual accuracy dipped when the figure faced left (45–90°: $M = .70$) and was higher when perspectives were aligned or inverted (0–45°: $M = .72$; 135–180°: $M = .81$; $\chi^2(3) = 22.99$, $p < .001$). A more pronounced U-shaped pattern was observed for o4-mini, which approached ceiling performance at low and high disparity for both visual (0–45°: $M = .99$; 135–180°: $M = .98$) and spatial questions (0–45°: $M = .99$; 135–180°: $M = .98$), with marked reductions at intermediate angles (visual: $M \approx .84$ –.86; spatial: $M \approx .85$ –.87; both $p < .001$).

In contrast, o3 showed the greatest robustness to both content and perspective variation. For this model, accuracy remained relatively stable across Angular Disparity bins for both visual (range $M = .78$ –.82; $\chi^2(3) = 4.19$, $p = .242$) and spatial

questions (range $M = .76$ –.79; $\chi^2(3) = 1.19$, $p = .756$), indicating no reliable effect of figure rotation.

Test 2: Level 2 VPT In Test 2, symbols were positioned in front of the figure and deviated slightly to the left or right, while remaining within the figure’s field of view. Visual questions probed how the symbol appeared from the figure’s perspective (e.g., whether it appeared as a 6 or a 9), whereas spatial questions asked whether the symbol was to the figure’s left or right. Figure 3 shows accuracy as a function of figure rotation for both question types, and Table 3 reports estimated marginal means across Angular Disparity bins along with simple main effects tests.

Table 3: Mean accuracy in Test 2 by Angular Disparity

Question	Model	0–45°	45–90°	90–135°	135–180°	χ^2	df	p
Visual	GPT-4o-mini	0.804	0.621	0.313	0.102	705.86	3	< .001
	GPT-4o	0.331	0.412	0.539	0.707	228.50	3	< .001
	o4-mini	0.462	0.408	0.650	0.875	368.29	3	< .001
	o3	0.650	0.548	0.495	0.656	55.70	3	< .001
Spatial	GPT-4o-mini	0.719	0.532	0.429	0.246	321.44	3	< .001
	GPT-4o	0.783	0.595	0.443	0.187	494.17	3	< .001
	o4-mini	0.868	0.668	0.631	0.802	134.96	3	< .001
	o3	0.611	0.565	0.499	0.659	42.35	3	< .001

Values are estimated marginal probabilities of correct responses.
Simple main effects test angular disparity within each Model $\times$ Question Type combination.

As in Test 1, we observed significant main effects of Model, Question Type, and Angular Disparity, as well as significant

interactions among these factors (all $p < .001$; Table 6), indicating that the impact of Angular Disparity varied across models and question types. For GPT-4o-mini, visual question accuracy declined sharply with increasing disparity, dropping from well above chance at low disparity (0–45°: $M = .80$) to near floor at high disparity (135–180°: $M = .10$; $\chi^2(3) = 705.86$, $p < .001$), consistent with systematic failures to adopt the figure’s visual perspective. A similar monotonic decline was observed for spatial questions, though performance differences were less severe (0–45°: $M = .72$; 135–180°: $M = .25$).

GPT-4o exhibited a qualitatively different pattern. For spatial questions, accuracy decreased steadily with Angular Disparity (0–45°: $M = .78$; 135–180°: $M = .19$; $\chi^2(3) = 494.17$, $p < .001$). In contrast, visual question performance was below chance when perspectives were shared (0–45°: $M = .33$) and increased when perspectives were inverted (135–180°: $M = .71$; $\chi^2(3) = 228.50$, $p < .001$), suggesting inconsistent or non-robust attempts to transform the symbol’s appearance.

The reasoning-oriented models o3 and o4-mini showed peak performance at low and high Angular Disparity for visual questions, with reduced accuracy at intermediate rotations. For o3, accuracy was higher when perspectives were aligned (0–45°: $M = .65$) or inverted (135–180°: $M = .66$) than at intermediate disparities (45–90°: $M = .55$; 90–135°: $M = .50$; $\chi^2(3) = 55.70$, $p < .001$). A similar U-shaped pattern was observed for o4-mini (0–45°: $M = .46$; 135–180°: $M = .88$), indicating that successful visual perspective-taking was largely confined to aligned or fully inverted viewpoints and did not generalize to intermediate rotations. For spatial questions, both models again showed U-shaped performance profiles, though accuracy was comparatively stable across disparities. This robustness was most pronounced for o4-mini, which maintained above-chance performance across all Angular Disparity bins (range $M = .63$ –.87; $\chi^2(3) = 134.96$, $p < .001$).

Test 3: Integrated visual and spatial PT In Test 3, two symbols were positioned within the figure’s field of view, slightly to the left and right. Unlike the previous tests, this condition involved a single visuospatial question probing how the symbol on the figure’s left or right side appeared to them, thereby jointly engaging visual and spatial perspective-taking.

Because only one visuospatial question was used, performance in Test 3 was analysed using an AI Model $\times$ Angular Disparity design. The analysis revealed a large main effect of Angular Disparity and a smaller but significant main effect of Model ($\chi^2(3) = 8.44$, $p = .038$; Table 7). A significant Model $\times$ Angular Disparity interaction was also observed, reflecting differences in the magnitude of performance decline across models.

Inspection of the estimated marginal means (Table 4) shows that all models exhibited steep declines in accuracy with increasing Angular Disparity. For example, GPT-4o-mini accuracy dropped from well above chance when perspectives were aligned (0–45°: $M = .79$) to near floor at high disparity (135–180°: $M = .01$; $\chi^2(3) = 689.37$, $p < .001$), with

a similar pattern observed for GPT-4o (0–45°: $M = .79$; 135–180°: $M = .02$). The reasoning-oriented models o3 and o4-mini were less severely affected, retaining higher accuracy at intermediate and high disparities (e.g., o3: 45–90° $M = .52$, 135–180° $M = .27$; o4-mini: 45–90° $M = .47$, 135–180° $M = .23$), though performance for both models nonetheless declined to approximately chance level (25%) at the largest disparities. Together, these results indicate a general failure to adopt visuospatial perspective in this task, even among models that showed relative robustness in earlier tests.

Table 4: Mean accuracy in Test 3 by Angular Disparity

Model	0–45°	45–90°	90–135°	135–180°	χ^2	df	p
GPT-4o-mini	0.790	0.332	0.087	0.011	689.37	3	< .001
GPT-4o	0.790	0.398	0.133	0.015	647.34	3	< .001
o4-mini	0.841	0.466	0.210	0.233	489.72	3	< .001
o3	0.810	0.523	0.227	0.272	415.62	3	< .001

Values are estimated marginal probabilities of correct responses.
Simple main effects test angular disparity within each Model.

Director Task

We analysed MLM performance on the *Director Task* using mixed-effects logistic regression models. Main and interaction effects were evaluated with Type III Wald χ^2 tests, and condition differences were examined via post-hoc comparisons of estimated marginal means.

Visual PT: Image vs ASCII We first assessed Level 2 *visual* perspective taking (visual PT) as a function of task format (image vs. ASCII). To isolate visual PT, we excluded spatial-different trials and restricted the analysis to trials presented from the director’s point of view. Mean accuracies are shown in Figure 4, with model summaries in Tables 8 and 9. Overall performance differed substantially across models ($\chi^2(1) = 498.04$, $p < .001$), and accuracy was lower in the image-based task ($M = 0.52$) than in the ASCII version ($M = 0.74$, $\chi^2(1) = 88.35$, $p < .001$), indicating a baseline cost of image processing.

Crucially, MLMs showed a pronounced visual PT deficit: accuracy dropped sharply from visual-shared trials ($M = 0.88$) to visual-different trials ($M = 0.35$; $\chi^2(3) = 141.27$, $p < .001$), and this deficit was robust across both formats. If anything, the visual-shared vs. visual-different gap was larger in ASCII ($\log(OR) = 26.88$) than in image trials ($\log(OR) = 7.00$), reflecting near-ceiling shared-perspective performance in the text version (ASCII: $M = 0.95$; Image: $M = 0.74$) rather than any selective disruption from image processing. Visual PT failures therefore cannot be attributed to image-specific perceptual demands.

The visual PT effect was large for all models (all $\log(OR) > 6$, all $p < .001$), though reasoning models showed more robust baselines across formats. Notably, o3 performed near ceiling on visual-shared trials ($M = 0.98$) yet showed a collapse in accuracy on visual-different trials ($M = 0.59$; $\log(OR) = 38.67$,

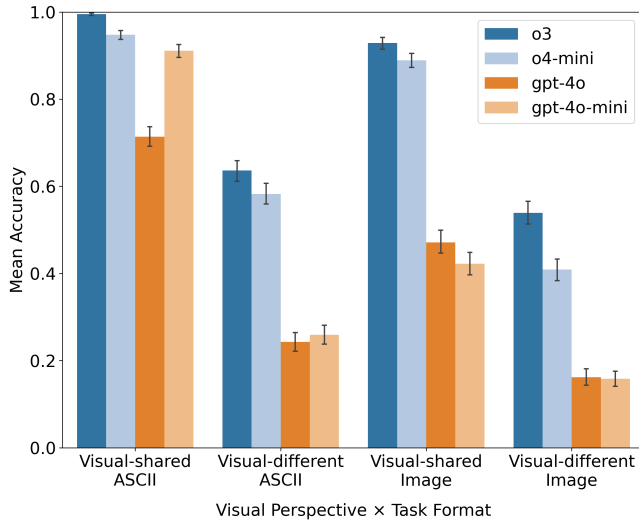


Figure 4: MLM mean accuracy in the *Director Task* for Visual Perspective conditions (Visual-shared and Visual-different) across Task Formats (ASCII and Image). Spatial-different and participant POV trials were excluded. Error bars are 95 percent confidence intervals.

$p < .001$), indicating a persistent failure to inhibit privileged visual information and adopt the director’s perspective.

Visual PT: Relative adjectives We next examined whether relative adjective type modulated visual perspective taking (Tables 10 and 11). We again excluded spatial-different trials and restricted analyses to director’s-POV trials, fitting an AI Model $\times$ Relative Adjective (None, Size, Spatial) $\times$ Visual Perspective (Visual-shared, Visual-different) model, with Task Format as a covariate. There was no main effect of Relative Adjective ($\chi^2(2) = 2.69$, $p = .260$), indicating that baseline performance did not differ across adjective conditions. The Relative Adjective $\times$ Visual Perspective interaction was weak ($\chi^2(2) = 6.25$, $p = .044$), and visual PT deficits were large under every adjective type.

Both the AI Model $\times$ Relative Adjective ($\chi^2(6) = 170.22$, $p < .001$) and three-way ($\chi^2(6) = 207.90$, $p < .001$) interactions were substantial. The two-way interaction reflects baseline capability differences: reasoning models (o3, o4-mini) showed consistently high visual-shared accuracy across adjective types (all $M > 0.89$), whereas GPT-4o variants performed less reliably. The three-way interaction reflects a sharper pattern in reasoning models, whose visual PT deficit was modest under simple referents (None: o3 $\log(OR) = 8.50$; o4-mini = 5.71) but expanded substantially when targets had to be disambiguated by size or spatial adjective (o3: $\log(OR) = 86.43$ and 28.17; o4-mini: $\log(OR) = 26.77$ and 24.23). VPT-related interference therefore scales with the referential demands of the trial, not just the visual perspective manipulation itself.

Spatial PT: Perspective reversals We next isolated Level 2 *spatial* perspective taking (spatial PT) by restricting analyses to visual-shared trials containing spatial adjectives, and fit an AI Model $\times$ Spatial Perspective (Shared: vertical adjective vs. Different: horizontal adjective) $\times$ Perspective Reversal (Participant POV vs. Director POV) model, controlling for Task Format (Tables 12 and 13). Vertical adjectives (“topmost”, “bottommost”) refer to the same items regardless of POV framing, so spatial PT is required only on horizontal-adjective trials from the director’s POV; the design therefore predicts a Spatial Perspective $\times$ Perspective Reversal interaction localised to that single cell.

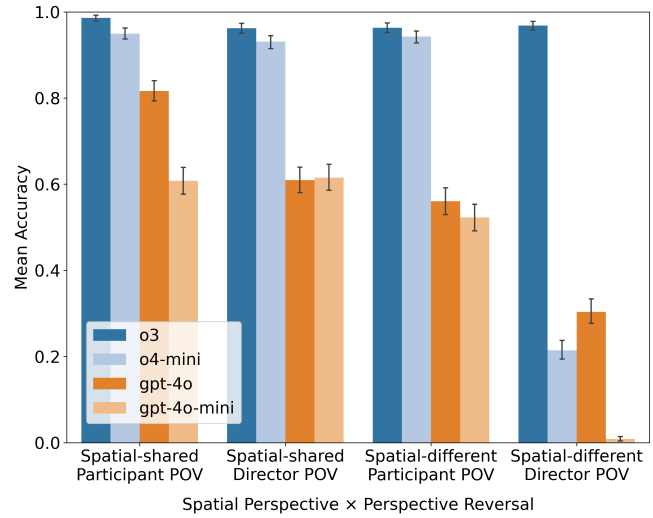


Figure 5: MLM mean accuracy in the *Director Task* on trials with spatial adjectives. Spatial perspective taking condition levels are Spatial-shared (Vertical: ‘topmost’/‘bottommost’) and Spatial-different (Horizontal: ‘leftmost’/‘rightmost’). Perspective Reversal condition levels are Participant point-of-view (POV; ‘from your point of view’) and Director POV (‘from my point of view’). The participant is required to adopt the director’s spatial perspective on spatial-different trials from the director’s POV only (rightmost grouping). Visual-different perspective-taking trials are not included. Error bars are 95 percent confidence intervals.

This pattern was confirmed. There was no main effect of Perspective Reversal ($\chi^2(1) = 0.02$, $p = .888$), but a strong Spatial Perspective $\times$ Perspective Reversal interaction ($\chi^2(1) = 120.93$, $p < .001$): performance was robust across the three control cells but dropped to $M = 0.28$ on horizontal trials from the director’s POV. A three-way interaction ($\chi^2(3) = 361.38$, $p < .001$) revealed model-specific patterns. GPT-4o-mini and o4-mini showed substantial spatial PT impairments in the critical cell ($\log(OR) = 224.27$ and 65.30, both $p < .001$). GPT-4o showed poorer performance on horizontal trials regardless of POV ($p < .001$ in both reversal conditions), suggesting a general difficulty with horizontal disambiguation rather than spatial PT specifically. Only o3 was unaffected by

the manipulation, performing near ceiling across all conditions (all $M > 0.96$; $\log(OR) = 0.89$, $p = .698$; see Figure 5).

We then investigated the interaction between visual and spatial Level 2 perspective taking demands by fitting an AI Model $\times$ Visual PT $\times$ Spatial PT mixed-effects model. The findings from this analysis are summarised in Tables 14, 15, and are described in the main text. Figure 6, displays the spatial PT and visual PT accuracy scores separately for image and ASCII tasks.

Finally, as the answers for the directors task are coordinates it is possible for us to plot the difference between the correct answer and the answer given for each answer a model got wrong but still gave a valid answer. We find that for samples that used "leftmost" or "rightmost" selectors each model, other than o3, was unable to inhibit its own perspective when the direction was given from the directors point of view. This can be seen clearly in figure 7 where we can see that for samples of the form "rightmost X from my perspective" models often selected items to the far right of the correct answer rather than realising that the director's right is their left. From this we can also see that for o3 the errors are clustered much closer to the correct answer.

Additionally, when we compare to the samples using "top-most" or "bottommost" this pattern does not appear (Figure 8), with most of the errors occurring to the left and right of the correct answer. This indicates some ability to understand the shared perspective when it does exist.

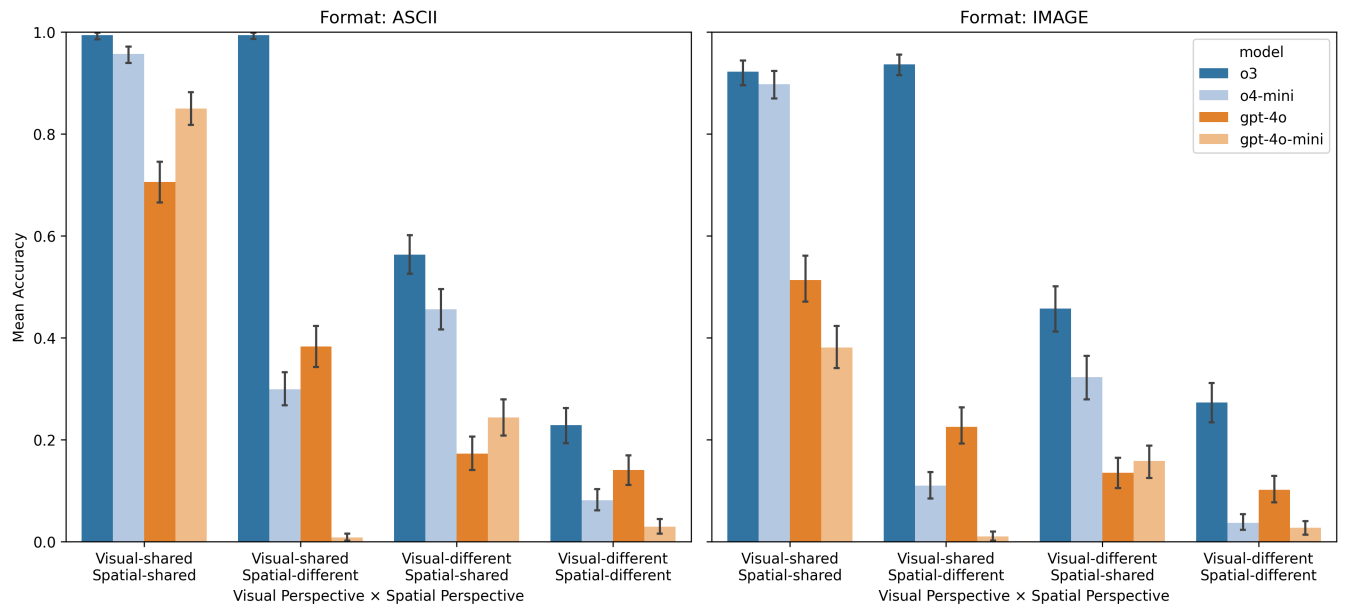


Figure 6: MLM mean accuracy in the *Director Task* on visual (occluded vs. no occluded alternatives) and spatial (horizontal vs. vertical adjectives) VPT trials. The data are plotted separately for Image and ASCII tasks, and include only spatial relative adjective trials from the director’s perspective. Error bars are 95 percent confidence intervals.

Table 5: Mixed effects logistic regression for Test 1: AI Model $\times$ Angular Disparity $\times$ Question Type

Term	Est./(χ^2)	SE/(df)	z	p
A. Model Coefficients				
Intercept	-0.466	0.085	-5.501	< 0.001***
Model $\times$ GPT-4o	1.430	0.116	12.339	< 0.001***
Model $\times$ o4-mini	4.988	0.316	15.780	< 0.001***
Model $\times$ o3	1.845	0.121	15.220	< 0.001***
Question Type $\times$ Spatial	0.787	0.112	7.052	< 0.001***
Angular Disparity $\times$ 67.5°	0.939	0.121	7.782	< 0.001***
Angular Disparity $\times$ 112.5°	1.352	0.124	10.933	< 0.001***
Angular Disparity $\times$ 157.5°	1.684	0.128	13.172	< 0.001***
Model $\times$ GPT-4o $\times$ Question Type $\times$ Spatial	-0.262	0.168	-1.565	0.117
Model $\times$ o4-mini $\times$ Question Type $\times$ Spatial	-0.327	0.500	-0.653	0.514
Model $\times$ o3 $\times$ Question Type $\times$ Spatial	-0.991	0.168	-5.900	< 0.001***
Model $\times$ GPT-4o $\times$ Angular Disparity $\times$ 67.5°	-1.071	0.164	-6.544	< 0.001***
Model $\times$ o4-mini $\times$ Angular Disparity $\times$ 67.5°	-3.781	0.340	-11.110	< 0.001***
Model $\times$ o3 $\times$ Angular Disparity $\times$ 67.5°	-0.778	0.173	-4.487	< 0.001***
Model $\times$ GPT-4o $\times$ Angular Disparity $\times$ 112.5°	-1.133	0.168	-6.733	< 0.001***
Model $\times$ o4-mini $\times$ Angular Disparity $\times$ 112.5°	-4.058	0.343	-11.846	< 0.001***
Model $\times$ o3 $\times$ Angular Disparity $\times$ 112.5°	-1.328	0.174	-7.632	< 0.001***
Model $\times$ GPT-4o $\times$ Angular Disparity $\times$ 157.5°	-1.219	0.174	-7.006	< 0.001***
Model $\times$ o4-mini $\times$ Angular Disparity $\times$ 157.5°	-2.116	0.416	-5.093	< 0.001***
Model $\times$ o3 $\times$ Angular Disparity $\times$ 157.5°	-1.805	0.176	-10.257	< 0.001***
Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 67.5°	-1.034	0.159	-6.518	< 0.001***
Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 112.5°	-1.599	0.161	-9.930	< 0.001***
Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 157.5°	-1.893	0.164	-11.515	< 0.001***
Model $\times$ GPT-4o $\times$ Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 67.5°	0.768	0.235	3.269	0.001**
Model $\times$ o4-mini $\times$ Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 67.5°	0.662	0.532	1.245	0.213
Model $\times$ o3 $\times$ Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 67.5°	1.000	0.241	4.150	< 0.001***
Model $\times$ GPT-4o $\times$ Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 112.5°	0.856	0.238	3.599	< 0.001***
Model $\times$ o4-mini $\times$ Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 112.5°	1.191	0.534	2.231	0.026*
Model $\times$ o3 $\times$ Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 112.5°	1.678	0.242	6.949	< 0.001***
Model $\times$ GPT-4o $\times$ Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 157.5°	1.190	0.244	4.876	< 0.001***
Model $\times$ o4-mini $\times$ Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 157.5°	1.200	0.618	1.941	0.052.
Model $\times$ o3 $\times$ Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 157.5°	2.049	0.243	8.431	< 0.001***
B. Type III Wald Tests				
Intercept	30.260	1		< 0.001***
Model	430.387	3		< 0.001***
Question Type	49.733	1		< 0.001***
Angular Disparity	203.713	3		< 0.001***
Model $\times$ Question Type	36.169	3		< 0.001***
Model $\times$ Angular Disparity	294.869	9		< 0.001***
Question Type $\times$ Angular Disparity	158.730	3		< 0.001***
Model $\times$ Question Type $\times$ Angular Disparity	87.884	9		< 0.001***

Panel A: Coefficient estimates with Wald z-tests for individual parameters.

Panel B: Type III Wald χ^2 tests for omnibus effects of factors.Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$ Random effect variances — image_id : 0.689

Table 6: Mixed effects logistic regression for Test 2: AI Model $\times$ Angular Disparity $\times$ Question Type

Term	Est./(χ^2)	SE/(df)	z	p
A. Model Coefficients				
Intercept	1.408	0.094	14.952	< 0.001***
Model $\times$ GPT-4o	-2.111	0.123	-17.158	< 0.001***
Model $\times$ o4-mini	-1.561	0.120	-12.988	< 0.001***
Model $\times$ o3	-0.787	0.122	-6.448	< 0.001***
Question Type $\times$ Spatial	-0.469	0.125	-3.752	< 0.001***
Angular Disparity $\times$ 67.5°	-0.913	0.122	-7.472	< 0.001***
Angular Disparity $\times$ 112.5°	-2.195	0.124	-17.709	< 0.001***
Angular Disparity $\times$ 157.5°	-3.585	0.149	-24.000	< 0.001***
Model $\times$ GPT-4o $\times$ Question Type $\times$ Spatial	2.453	0.174	14.122	< 0.001***
Model $\times$ o4-mini $\times$ Question Type $\times$ Spatial	2.505	0.182	13.728	< 0.001***
Model $\times$ o3 $\times$ Question Type $\times$ Spatial	0.300	0.166	1.804	0.071.
Model $\times$ GPT-4o $\times$ Angular Disparity $\times$ 67.5°	1.262	0.164	7.697	< 0.001***
Model $\times$ o4-mini $\times$ Angular Disparity $\times$ 67.5°	0.695	0.162	4.289	< 0.001***
Model $\times$ o3 $\times$ Angular Disparity $\times$ 67.5°	0.485	0.163	2.973	0.003**
Model $\times$ GPT-4o $\times$ Angular Disparity $\times$ 112.5°	3.055	0.165	18.557	< 0.001***
Model $\times$ o4-mini $\times$ Angular Disparity $\times$ 112.5°	2.968	0.164	18.093	< 0.001***
Model $\times$ o3 $\times$ Angular Disparity $\times$ 112.5°	1.555	0.164	9.502	< 0.001***
Model $\times$ GPT-4o $\times$ Angular Disparity $\times$ 157.5°	5.167	0.186	27.761	< 0.001***
Model $\times$ o4-mini $\times$ Angular Disparity $\times$ 157.5°	5.685	0.198	28.737	< 0.001***
Model $\times$ o3 $\times$ Angular Disparity $\times$ 157.5°	3.609	0.184	19.628	< 0.001***
Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 67.5°	0.104	0.165	0.630	0.529
Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 112.5°	0.971	0.166	5.846	< 0.001***
Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 157.5°	1.527	0.188	8.102	< 0.001***
Model $\times$ GPT-4o $\times$ Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 67.5°	-1.352	0.231	-5.849	< 0.001***
Model $\times$ o4-mini $\times$ Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 67.5°	-1.071	0.239	-4.485	< 0.001***
Model $\times$ o3 $\times$ Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 67.5°	0.133	0.225	0.592	0.554
Model $\times$ GPT-4o $\times$ Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 112.5°	-3.342	0.231	-14.471	< 0.001***
Model $\times$ o4-mini $\times$ Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 112.5°	-3.090	0.239	-12.921	< 0.001***
Model $\times$ o3 $\times$ Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 112.5°	-0.786	0.225	-3.491	< 0.001***
Model $\times$ GPT-4o $\times$ Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 157.5°	-5.860	0.254	-23.077	< 0.001***
Model $\times$ o4-mini $\times$ Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 157.5°	-4.112	0.269	-15.314	< 0.001***
Model $\times$ o3 $\times$ Question Type $\times$ Spatial $\times$ Angular Disparity $\times$ 157.5°	-1.342	0.242	-5.539	< 0.001***
B. Type III Wald Tests				
Intercept	223.561	1		< 0.001***
Model	344.468	3		< 0.001***
Question Type	14.079	1		< 0.001***
Angular Disparity	705.865	3		< 0.001***
Model $\times$ Question Type	362.226	3		< 0.001***
Model $\times$ Angular Disparity	1259.117	9		< 0.001***
Question Type $\times$ Angular Disparity	98.757	3		< 0.001***
Model $\times$ Question Type $\times$ Angular Disparity	814.383	9		< 0.001***

Panel A: Coefficient estimates with Wald z-tests for individual parameters.

Panel B: Type III Wald χ^2 tests for omnibus effects of factors.

Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$

Random effect variances — image_i d : 0.076

Table 7: Mixed effects logistic regression for Test 3: AI Model $\times$ Angular Disparity

Term	Est./(χ^2)	SE/(df)	z	p
A. Model Coefficients				
Intercept	1.323	0.101	13.060	< 0.001***
Model $\times$ GPT-4o	-0.000	0.130	-0.000	1.000
Model $\times$ o4-mini	0.341	0.134	2.536	0.011*
Model $\times$ o3	0.129	0.131	0.984	0.325
Angular Disparity $\times$ 67.5°	-2.023	0.138	-14.659	< 0.001***
Angular Disparity $\times$ 112.5°	-3.679	0.167	-22.085	< 0.001***
Angular Disparity $\times$ 157.5°	-5.801	0.307	-18.924	< 0.001***
Model $\times$ GPT-4o $\times$ Angular Disparity $\times$ 67.5°	0.285	0.175	1.634	0.102
Model $\times$ o4-mini $\times$ Angular Disparity $\times$ 67.5°	0.221	0.178	1.243	0.214
Model $\times$ o3 $\times$ Angular Disparity $\times$ 67.5°	0.661	0.176	3.763	< 0.001***
Model $\times$ GPT-4o $\times$ Angular Disparity $\times$ 112.5°	0.478	0.204	2.338	0.019*
Model $\times$ o4-mini $\times$ Angular Disparity $\times$ 112.5°	0.688	0.202	3.407	< 0.001***
Model $\times$ o3 $\times$ Angular Disparity $\times$ 112.5°	1.001	0.199	5.017	< 0.001***
Model $\times$ GPT-4o $\times$ Angular Disparity $\times$ 157.5°	0.278	0.397	0.701	0.483
Model $\times$ o4-mini $\times$ Angular Disparity $\times$ 157.5°	2.947	0.325	9.055	< 0.001***
Model $\times$ o3 $\times$ Angular Disparity $\times$ 157.5°	3.363	0.324	10.384	< 0.001***
B. Type III Wald Tests				
Intercept	170.575	1		< 0.001***
Model	8.444	3		0.038*
Angular Disparity	689.370	3		< 0.001***
Model $\times$ Angular Disparity	210.335	9		< 0.001***

Panel A: Coefficient estimates with Wald z-tests for individual parameters.

Panel B: Type III Wald χ^2 tests for omnibus effects of factors.

Significance codes: *** p < 0.001, ** p < 0.01, * p < 0.05, . p < 0.1

Random effect variances — image; d : 1.071

Table 8: Mixed effects logistic regression: AI Model $\times$ Visual PT $\times$ Task Format

Term	Est./(χ^2)	SE/(df)	z	p
A. Model Coefficients				
Intercept	2.427	0.214	11.352	< 0.001***
Model: GPT-4o	-1.455	0.109	-13.394	< 0.001***
Model: o4-mini	0.585	0.139	4.222	< 0.001***
Model: o3	3.031	0.346	8.752	< 0.001***
Format: Image	-2.748	0.292	-9.400	< 0.001***
Visual PT: Different	-3.490	0.294	-11.886	< 0.001***
Model: GPT-4o $\times$ Format: Image	1.657	0.131	12.610	< 0.001***
Model: o4-mini $\times$ Format: Image	1.831	0.170	10.801	< 0.001***
Model: o3 $\times$ Format: Image	-0.124	0.364	-0.339	0.735
Model: GPT-4o $\times$ Visual PT: Different	1.368	0.138	9.918	< 0.001***
Model: o4-mini $\times$ Visual PT: Different	0.822	0.158	5.196	< 0.001***
Model: o3 $\times$ Visual PT: Different	-1.394	0.355	-3.930	< 0.001***
Format: Image $\times$ Visual PT: Different	2.132	0.410	5.200	< 0.001***
Model: GPT-4o $\times$ Format: Image $\times$ Visual PT: Different	-1.545	0.186	-8.314	< 0.001***
Model: o4-mini $\times$ Format: Image $\times$ Visual PT: Different	-1.928	0.206	-9.358	< 0.001***
Model: o3 $\times$ Format: Image $\times$ Visual PT: Different	0.328	0.383	0.856	0.392
B. Type III Wald Tests				
Intercept	128.877	1		< 0.001***
Model	498.038	3		< 0.001***
Format	88.354	1		< 0.001***
Visual PT	141.269	1		< 0.001***
Model $\times$ Format	200.406	3		< 0.001***
Model $\times$ Visual PT	141.104	3		< 0.001***
Format $\times$ Visual PT	27.044	1		< 0.001***
Model $\times$ Format $\times$ Visual PT	121.048	3		< 0.001***

Panel A: Coefficient estimates with Wald z-tests for individual parameters

Panel B: Type III Wald χ^2 tests for omnibus effects of factors

Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$

Random effect variance (Trial ID): 0.871

Table 9: Pairwise comparisons with estimated marginal means: AI Model $\times$ Visual PT $\times$ Task Format

Contrast	Context	Estimated marginal means		Estimate	SE	z	p
A. Perspective × Format							
Shared vs. Different	ASCII	Shared: 0.951	Different: 0.420	26.881	7.766	11.393	< 0.001***
Shared vs. Different	Image	Shared: 0.743	Different: 0.292	6.999	1.934	7.041	< 0.001***
B. Format × Perspective							
ASCII vs. Image	Visual-shared	ASCII: 0.951	Image: 0.743	6.730	1.952	6.574	< 0.001***
ASCII vs. Image	Visual-different	ASCII: 0.420	Image: 0.292	1.752	0.482	2.038	0.042*
C. Perspective × Model							
Shared vs. Different	gpt-4o-mini	Shared: 0.741	Different: 0.202	11.295	2.315	11.828	< 0.001***
Shared vs. Different	gpt-4o	Shared: 0.605	Different: 0.197	6.225	1.256	9.060	< 0.001***
Shared vs. Different	o4-mini	Shared: 0.928	Different: 0.497	13.021	2.693	12.408	< 0.001***
Shared vs. Different	o3	Shared: 0.982	Different: 0.591	38.665	10.143	13.932	< 0.001***
D. Format × Model							
ASCII vs. Image	gpt-4o-mini	ASCII: 0.664	Image: 0.269	5.375	1.102	8.204	< 0.001***
ASCII vs. Image	gpt-4o	ASCII: 0.478	Image: 0.292	2.220	0.448	3.951	< 0.001***
ASCII vs. Image	o4-mini	ASCII: 0.843	Image: 0.703	2.257	0.467	3.936	< 0.001***
ASCII vs. Image	o3	ASCII: 0.953	Image: 0.798	5.162	1.354	6.257	< 0.001***

Estimate is the the log-odds ratio between conditions.

P-values adjusted using the Bonferroni method within each family of comparisons.

Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, . $p < 0.1$

Table 10: Mixed effects logistic regression: AI Model $\times$ Visual PT $\times$ Relative Adjective

Term	Est./(χ^2)	SE/(df)	z	p
A. Model Coefficients				
Intercept	1.618	0.249	6.501	< 0.001***
Model: GPT-4o	-1.380	0.103	-13.341	< 0.001***
Model: o4-mini	1.106	0.121	9.103	< 0.001***
Model: o3	2.055	0.158	13.034	< 0.001***
Rel. Adj: Size	-0.119	0.327	-0.365	0.715
Rel. Adj: Spatial	-0.510	0.326	-1.567	0.117
Visual PT: Different	-2.900	0.331	-8.757	< 0.001***
Format: Image	-1.153	0.182	-6.342	< 0.001***
Model: GPT-4o $\times$ Rel. Adj: Size	1.765	0.149	11.821	< 0.001***
Model: o4-mini $\times$ Rel. Adj: Size	1.297	0.199	6.507	< 0.001***
Model: o3 $\times$ Rel. Adj: Size	1.621	0.302	5.372	< 0.001***
Model: GPT-4o $\times$ Rel. Adj: Spatial	1.351	0.142	9.508	< 0.001***
Model: o4-mini $\times$ Rel. Adj: Spatial	1.097	0.184	5.964	< 0.001***
Model: o3 $\times$ Rel. Adj: Spatial	0.791	0.233	3.391	< 0.001***
Model: GPT-4o $\times$ Visual PT: Different	1.621	0.162	9.989	< 0.001***
Model: o4-mini $\times$ Visual PT: Different	1.157	0.165	7.015	< 0.001***
Model: o3 $\times$ Visual PT: Different	0.760	0.195	3.892	< 0.001***
Rel. Adj: Size $\times$ Visual PT: Different	1.043	0.465	2.244	0.025*
Rel. Adj: Spatial $\times$ Visual PT: Different	0.970	0.465	2.087	0.037*
Model: GPT-4o $\times$ Rel. Adj: Size $\times$ Visual PT: Different	-2.016	0.219	-9.189	< 0.001***
Model: o4-mini $\times$ Rel. Adj: Size $\times$ Visual PT: Different	-2.588	0.247	-10.475	< 0.001***
Model: o3 $\times$ Rel. Adj: Size $\times$ Visual PT: Different	-3.363	0.336	-9.994	< 0.001***
Model: GPT-4o $\times$ Rel. Adj: Spatial $\times$ Visual PT: Different	-1.923	0.223	-8.622	< 0.001***
Model: o4-mini $\times$ Rel. Adj: Spatial $\times$ Visual PT: Different	-2.415	0.237	-10.179	< 0.001***
Model: o3 $\times$ Rel. Adj: Spatial $\times$ Visual PT: Different	-2.169	0.278	-7.790	< 0.001***
B. Type III Wald Tests				
Intercept	42.268	1		< 0.001***
Model	701.100	3		< 0.001***
Rel. Adj	2.693	2		0.260
Visual PT	76.687	1		< 0.001***
Format	40.226	1		< 0.001***
Model $\times$ Rel. Adj	170.216	6		< 0.001***
Model $\times$ Visual PT	105.717	3		< 0.001***
Rel. Adj $\times$ Visual PT	6.247	2		0.044*
Model $\times$ Rel. Adj $\times$ Visual PT	207.900	6		< 0.001***

Panel A: Coefficient estimates with Wald z-tests for individual parameters

Panel B: Type III Wald χ^2 tests for omnibus effects of factors

Significance codes: *** p < 0.001, ** p < 0.01, * p < 0.05, . p < 0.1

Random effect variance (Trial ID): 0.394

Table 11: Pairwise comparisons with estimated marginal means: AI Model $\times$ Visual PT $\times$ Relative Adjective

Contrast	Context	Estimated marginal means		Estimate	SE	z	p
A. GPT-4o-mini							
Shared vs. Different	None	Shared: 0.739	Different: 0.135	18.178	6.020	8.757	< 0.001***
Shared vs. Different	Size	Shared: 0.715	Different: 0.282	6.403	2.090	5.688	< 0.001***
Shared vs. Different	Spatial	Shared: 0.630	Different: 0.198	6.889	2.248	5.915	< 0.001***
B. GPT-4o							
Shared vs. Different	None	Shared: 0.416	Different: 0.166	3.594	1.178	3.903	< 0.001***
Shared vs. Different	Size	Shared: 0.787	Different: 0.280	9.504	3.115	6.870	< 0.001***
Shared vs. Different	Spatial	Shared: 0.623	Different: 0.151	9.318	3.060	6.795	< 0.001***
C. o4-mini							
Shared vs. Different	None	Shared: 0.895	Different: 0.600	5.713	1.885	5.281	< 0.001***
Shared vs. Different	Size	Shared: 0.965	Different: 0.509	26.765	9.269	9.492	< 0.001***
Shared vs. Different	Spatial	Shared: 0.939	Different: 0.389	24.228	8.172	9.451	< 0.001***
D. o3							
Shared vs. Different	None	Shared: 0.957	Different: 0.722	8.501	2.943	6.182	< 0.001***
Shared vs. Different	Size	Shared: 0.990	Different: 0.535	86.428	34.709	11.104	< 0.001***
Shared vs. Different	Spatial	Shared: 0.967	Different: 0.510	28.174	9.923	9.479	< 0.001***

Estimate is the the log-odds ratio between conditions.

P-values adjusted using the Bonferroni method within each family of comparisons.

Significance codes: *** p < 0.001, ** p < 0.01, * p < 0.05, . p < 0.1

Table 12: Mixed effects logistic regression: AI Model $\times$ Spatial PT $\times$ Perspective Reversal

Term	Est./(χ^2)	SE/(df)	z	p
A. Model Coefficients				
Intercept	1.310	0.173	7.566	< 0.001***
Model: GPT-4o	1.261	0.115	10.942	< 0.001***
Model: o4-mini	2.705	0.159	17.053	< 0.001***
Model: o3	4.041	0.265	15.260	< 0.001***
Spatial PT: Different	-0.461	0.218	-2.112	0.035*
Persp. Reversal: Director POV	-0.031	0.219	-0.141	0.888
Format: Image	-1.493	0.146	-10.256	< 0.001***
Model: GPT-4o $\times$ Spatial PT: Different	-1.092	0.151	-7.248	< 0.001***
Model: o4-mini $\times$ Spatial PT: Different	0.116	0.218	0.533	0.594
Model: o3 $\times$ Spatial PT: Different	-0.751	0.319	-2.358	0.018*
Model: GPT-4o $\times$ Persp. Reversal: Director POV	-1.290	0.151	-8.540	< 0.001***
Model: o4-mini $\times$ Persp. Reversal: Director POV	-0.500	0.210	-2.375	0.018*
Model: o3 $\times$ Persp. Reversal: Director POV	-1.193	0.316	-3.778	< 0.001***
Spatial PT: Different $\times$ Persp. Reversal: Director POV	-4.952	0.450	-10.997	< 0.001***
Model: GPT-4o $\times$ Spatial PT: Different $\times$ Persp. Reversal: Director POV	5.113	0.387	13.226	< 0.001***
Model: o4-mini $\times$ Spatial PT: Different $\times$ Persp. Reversal: Director POV	1.118	0.429	2.603	0.009**
Model: o3 $\times$ Spatial PT: Different $\times$ Persp. Reversal: Director POV	6.281	0.523	12.021	< 0.001***
B. Type III Wald Tests				
Intercept	57.241	1		< 0.001***
Model	472.449	3		< 0.001***
Spatial PT	4.461	1		0.035*
Persp. Reversal	0.020	1		0.888
Format	105.178	1		< 0.001***
Model $\times$ Spatial PT	62.842	3		< 0.001***
Model $\times$ Persp. Reversal	77.080	3		< 0.001***
Spatial PT $\times$ Persp. Reversal	120.928	1		< 0.001***
Model $\times$ Spatial PT $\times$ Persp. Reversal	361.378	3		< 0.001***

Panel A: Coefficient estimates with Wald z-tests for individual parameters

Panel B: Type III Wald χ^2 tests for omnibus effects of factors

Significance codes: *** p < 0.001, ** p < 0.01, * p < 0.05, . p < 0.1

Random effect variance (Trial ID): 2.073

Table 13: Pairwise comparisons with estimated marginal means: AI Model $\times$ Spatial PT $\times$ Perspective Reversal

Contrast	Context	Estimated marginal means		Estimate	SE	z	p
A. GPT-4o-mini							
Shared vs. Different	Participant POV	Shared: 0.637	Different: 0.526	1.586	0.346	2.112	0.035*
Shared vs. Different	Director POV	Shared: 0.630	Different: 0.008	224.268	88.320	13.745	< 0.001***
B. GPT-4o							
Shared vs. Different	Participant POV	Shared: 0.861	Different: 0.568	4.724	1.067	6.873	< 0.001***
Shared vs. Different	Director POV	Shared: 0.623	Different: 0.292	4.023	0.873	6.411	< 0.001***
C. o4-mini							
Shared vs. Different	Participant POV	Shared: 0.963	Different: 0.949	1.411	0.390	1.248	0.212
Shared vs. Different	Director POV	Shared: 0.939	Different: 0.191	65.301	15.693	17.389	< 0.001***
D. o3							
Shared vs. Different	Participant POV	Shared: 0.990	Different: 0.967	3.361	1.212	3.361	< 0.001***
Shared vs. Different	Director POV	Shared: 0.967	Different: 0.971	0.890	0.268	-0.388	0.698

Estimate is the the log-odds ratio between conditions.

P-values adjusted using the Bonferroni method within each family of comparisons.

Significance codes: *** p < 0.001, ** p < 0.01, * p < 0.05, . p < 0.1

Table 14: Mixed effects logistic regression: AI Model $\times$ Visual PT $\times$ Spatial PT

Term	Est./(χ^2)	SE/(df)	z	p
A. Model Coefficients				
Intercept	0.909	0.210	4.325	< 0.001***
Model: GPT-4o	-0.028	0.097	-0.292	0.770
Model: o4-mini	2.199	0.138	15.918	< 0.001***
Model: o3	2.841	0.172	16.537	< 0.001***
Visual PT: Different	-1.927	0.270	-7.135	< 0.001***
Spatial PT: Different	-5.391	0.424	-12.727	< 0.001***
Format: Image	-0.760	0.182	-4.186	< 0.001***
Model: GPT-4o $\times$ Visual PT: Different	-0.302	0.153	-1.976	0.048*
Model: o4-mini $\times$ Visual PT: Different	-1.253	0.170	-7.355	< 0.001***
Model: o3 $\times$ Visual PT: Different	-1.404	0.198	-7.080	< 0.001***
Model: GPT-4o $\times$ Spatial PT: Different	4.011	0.356	11.271	< 0.001***
Model: o4-mini $\times$ Spatial PT: Different	1.238	0.370	3.350	< 0.001***
Model: o3 $\times$ Spatial PT: Different	5.506	0.414	13.300	< 0.001***
Visual PT: Different $\times$ Spatial PT: Different	3.231	0.532	6.073	< 0.001***
Model: GPT-4o $\times$ Visual PT: Different $\times$ Spatial PT: Different	-2.110	0.430	-4.905	< 0.001***
Model: o4-mini $\times$ Visual PT: Different $\times$ Spatial PT: Different	-1.366	0.444	-3.080	0.002**
Model: o3 $\times$ Visual PT: Different $\times$ Spatial PT: Different	-4.506	0.470	-9.580	< 0.001***
B. Type III Wald Tests				
Intercept	18.703	1		< 0.001***
Model	529.906	3		< 0.001***
Visual PT	50.906	1		< 0.001***
Spatial PT	161.973	1		< 0.001***
Format	17.522	1		< 0.001***
Model $\times$ Visual PT	85.176	3		< 0.001***
Model $\times$ Spatial PT	449.090	3		< 0.001***
Visual PT $\times$ Spatial PT	36.886	1		< 0.001***
Model $\times$ Visual PT $\times$ Spatial PT	134.516	3		< 0.001***

Panel A: Coefficient estimates with Wald z-tests for individual parameters

Panel B: Type III Wald χ^2 tests for omnibus effects of factors

Significance codes: *** p < 0.001, ** p < 0.01, * p < 0.05, . p < 0.1

Random effect variance (Trial ID): 0.075

Table 15: Pairwise comparisons with estimated marginal means: AI Model $\times$ Visual PT $\times$ Spatial PT

Contrast	Context	Estimated marginal means		Estimate	SE	z	p
A. Visual × Spatial							
Visual-shared vs. Visual-different	Spatial-shared	Shared: 0.856	Different: 0.292	14.390	3.710	10.344	< 0.001***
Visual-shared vs. Visual-different	Spatial-different	Shared: 0.285	Different: 0.087	4.183	1.148	5.212	< 0.001***
B. Spatial × Visual							
Spatial-shared vs. Spatial-different	Visual-shared	Shared: 0.856	Different: 0.285	14.914	4.073	9.894	< 0.001***
Spatial-shared vs. Spatial-different	Visual-different	Shared: 0.292	Different: 0.087	4.335	1.124	5.656	< 0.001***
C. Visual × Model							
Visual-shared vs. Visual-different	gpt-4o-mini	Shared: 0.103	Different: 0.077	1.365	0.363	1.171	0.242
Visual-shared vs. Visual-different	gpt-4o	Shared: 0.453	Different: 0.135	5.303	1.027	8.610	< 0.001***
Visual-shared vs. Visual-different	o4-mini	Shared: 0.657	Different: 0.168	9.465	1.908	11.148	< 0.001***
Visual-shared vs. Visual-different	o3	Shared: 0.969	Different: 0.368	52.875	11.386	18.426	< 0.001***
D. Spatial × Model							
Spatial-shared vs. Spatial-different	gpt-4o-mini	Shared: 0.393	Different: 0.015	43.618	11.602	14.194	< 0.001***
Spatial-shared vs. Spatial-different	gpt-4o	Shared: 0.351	Different: 0.193	2.270	0.440	4.230	< 0.001***
Spatial-shared vs. Spatial-different	o4-mini	Shared: 0.757	Different: 0.111	25.043	5.050	15.972	< 0.001***
Spatial-shared vs. Spatial-different	o3	Shared: 0.846	Different: 0.765	1.686	0.363	2.426	0.015*

Estimate is the the log-odds ratio between conditions.

P-values adjusted using the Bonferroni method within each family of comparisons.

Significance codes: *** p < 0.001, ** p < 0.01, * p < 0.05, . p < 0.1

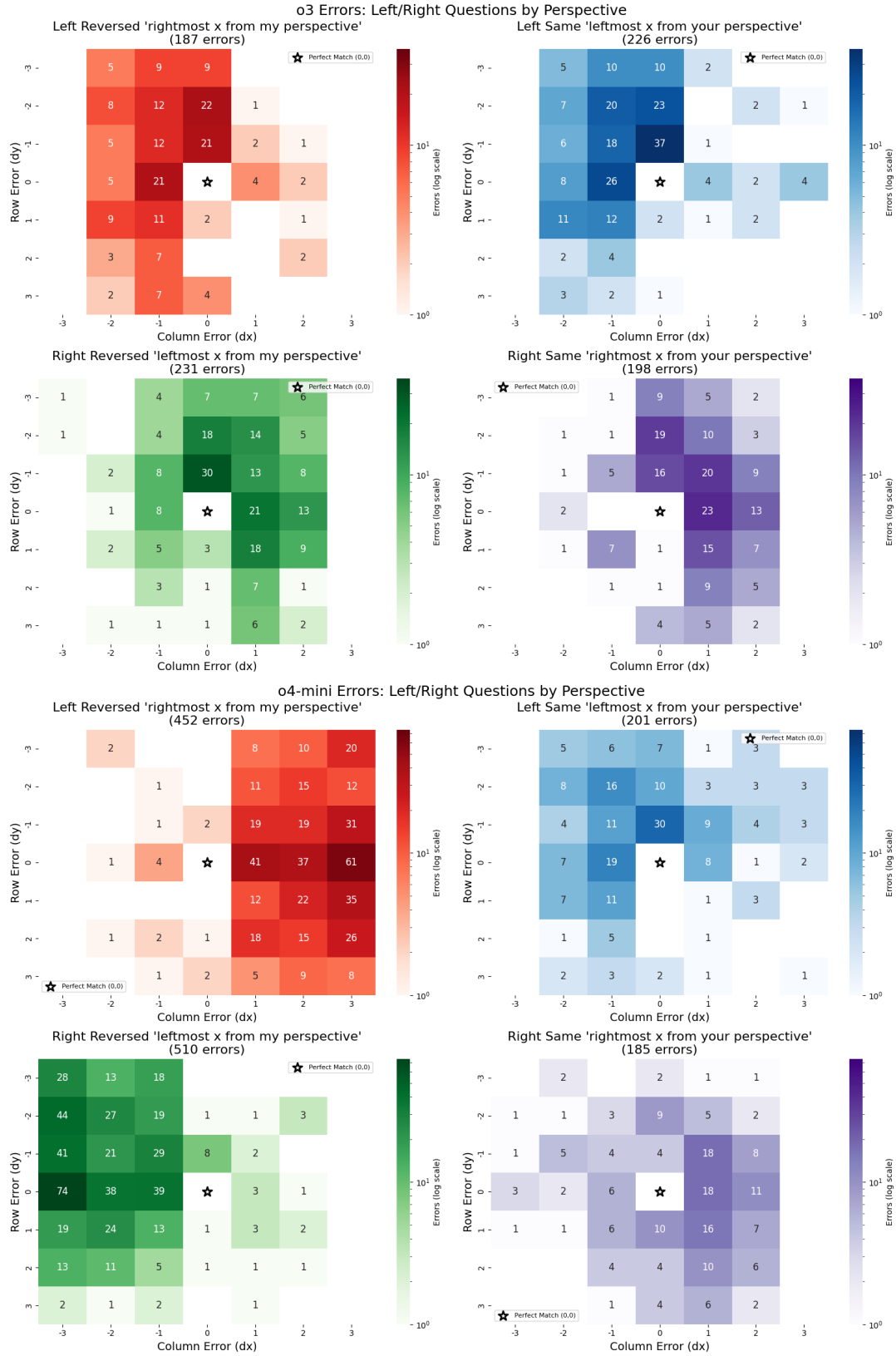


Figure 7: Relative position of each error made by the models for questions using "leftmost" or "rightmost" selectors split by model and then by perspective.

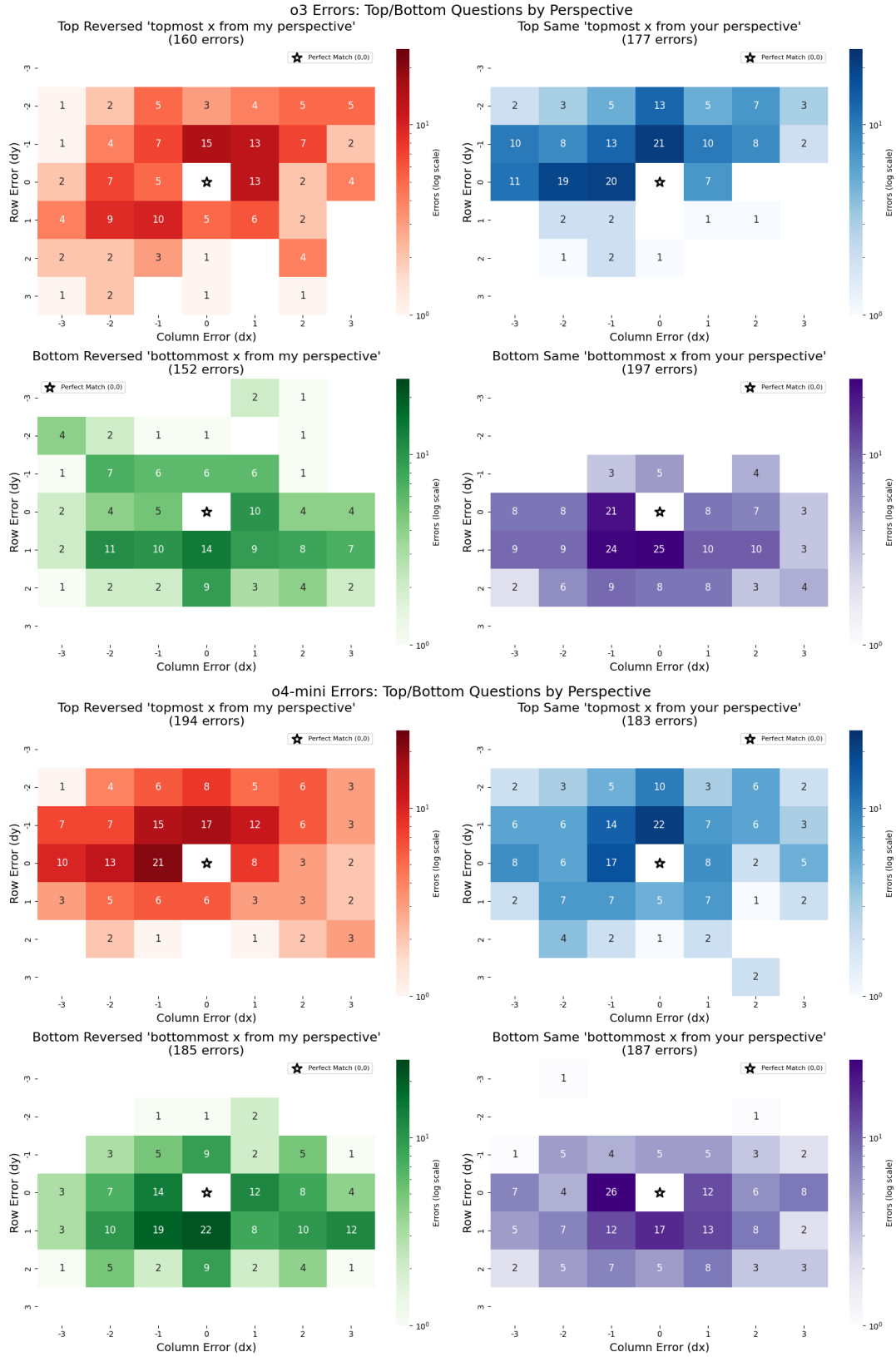


Figure 8: Relative position of each error made by the models for questions using "topmost" or "bottommost" selectors split by model and then by perspective.

Human Baseline

To establish the behavioural signature of intact visuospatial perspective taking, we collected a human baseline on the same Rotating Figure Task battery presented to the MLMs. Human performance provides the reference against which the models' failure profiles are interpreted, and in particular reveals the graded cost of angular disparity – in both accuracy and response time – that characterises genuine Level 2 perspective transformation (Surtees et al., 2013).

Participants and procedure

This experiment was approved by the Research Ethics Committee at the Department of Psychology University of Cambridge. Thirty adults ($N = 30$) recruited via *Prolific* completed an online version of the Rotating Figure Task administered through *Gorilla* (<https://gorilla.sc>). After providing informed consent, each participant responded to a balanced sample of 48 trials per condition (5 conditions, 240 trials in total) drawn from the same procedurally generated battery used to evaluate the MLMs, organised into blocks by stimulus set and question type. On each trial participants viewed a single stimulus image and answered the same question posed to the models: visual and spatial variants for Test 1–2 conditions, and a single visuospatial question for Test 3 (e.g., “*What number or letter can the person see on their left/right side?*”). Both response accuracy and reaction time were recorded.

Analysis

Accuracy was analysed with the same mixed-effects logistic regression approach used for the MLMs, modelling Angular Disparity (and its interaction with Question Type for Tests 1–2) with by-participant random intercepts. Reaction times (correct trials with a valid, non-zero response time) were analysed with linear mixed models fit to log-transformed RT, with by-participant and by-image random intercepts, and are reported here as the exponentiated mean log-RT. Estimated marginal means and simple main-effect tests of Angular Disparity are given in Table 16 (accuracy) and Table 17 (reaction time). Humans were additionally entered alongside the MLMs in the combined model reported in the main text.

Accuracy

Participants performed near ceiling on Test 1 (Level 1 VPT), with accuracy invariant to angular disparity for both visual (M range .97–.98; $\chi^2(3) = 1.36$, $p = .715$) and spatial questions (.97–.99; $\chi^2(3) = 5.08$, $p = .166$), consistent with the automatic character of Level 1 visibility judgements. The Level 2 conditions instead produced the hallmark graded cost: accuracy declined monotonically with disparity in Test 2 (visual .95 to .77; spatial .99 to .90) and Test 3 (visuospatial .94 to .73; all $p < .001$; Table 16). This graded decline is the pattern expected if adopting a rotated viewpoint becomes progressively more demanding as angular disparity grows.

Table 16: Human accuracy by Angular Disparity across test conditions

Test	Question	0–45°	45–90°	90–135°	135–180°	χ^2	df	p
Test 1	Visual	0.971	0.975	0.965	0.977	1.36	3	0.715
	Spatial	0.984	0.973	0.992	0.984	5.08	3	0.166
Test 2	Visual	0.952	0.902	0.851	0.771	38.31	3	< .001
	Spatial	0.991	0.935	0.934	0.904	33.55	3	< .001
Test 3	Visuospatial	0.938	0.891	0.805	0.727	31.89	3	< .001

Values are estimated marginal probabilities of correct responses ($N = 30$).

$\chi^2(3)$ tests the simple main effect of angular disparity within each condition.

Reaction time

The reaction-time data followed the same pattern as accuracy. Level 1 responses were fast (≈ 1000 ms) and unaffected by angular disparity (both $p > .29$), consistent with rapid, near-automatic visibility computation. Level 2 responses were substantially slower – roughly 1.2–2.1 seconds in Test 2 and 2.9–3.4 seconds in Test 3 – and increased with angular disparity: for Test 2 spatial (1171 to 1592 ms, $\chi^2(3) = 60.83$, $p < .001$) and visual questions (1812 to 2094 ms, $\chi^2(3) = 9.14$, $p = .028$), and also for the visuospatial question in Test 3 (2889 to 3423 ms, $\chi^2(3) = 25.91$, $p < .001$; Table 17). A response-time cost that scales with the required transformation is the classic behavioural signature of mental rotation (Surtees et al., 2013): humans can often resolve non-shared perspectives accurately, but pay a graded temporal cost to do so.

Table 17: Human reaction time (geometric mean, ms) by Angular Disparity across test conditions

Test	Question	0–45°	45–90°	90–135°	135–180°	χ^2	df	p
Test 1	Visual	1046	1012	1001	997	2.69	3	0.442
	Spatial	1006	1011	960	975	3.68	3	0.299
Test 2	Visual	1812	1934	1984	2094	9.14	3	0.028
	Spatial	1171	1215	1469	1592	60.83	3	< .001
Test 3	Visuospatial	2889	3312	3300	3423	25.91	3	< .001

Values are geometric mean reaction times (ms), back-transformed from log scale ($N = 30$).

$\chi^2(3)$ tests the simple main effect of angular disparity within each condition.

Comparison with MLMs

Figure 9 compares human and MLM accuracy across test conditions and angular disparity. Humans (dashed) remain at ceiling on Test 1 and show a gradual decline with disparity on Tests 2–3. By contrast, the models exhibit qualitatively non-human profiles: reduced accuracy or inverted trajectories at Level 1, U-shaped or inverted patterns for the reasoning models at Level 2, and steep performance collapses under the integrated demands of Test 3. No model reproduces the human pattern of accurate, graded performance or the disparity-dependent response-time cost that, in humans, is the hallmark of genuine perspective transformation. The human baseline therefore serves as both a performance ceiling and a mechanistic benchmark against which the models consistently diverge.

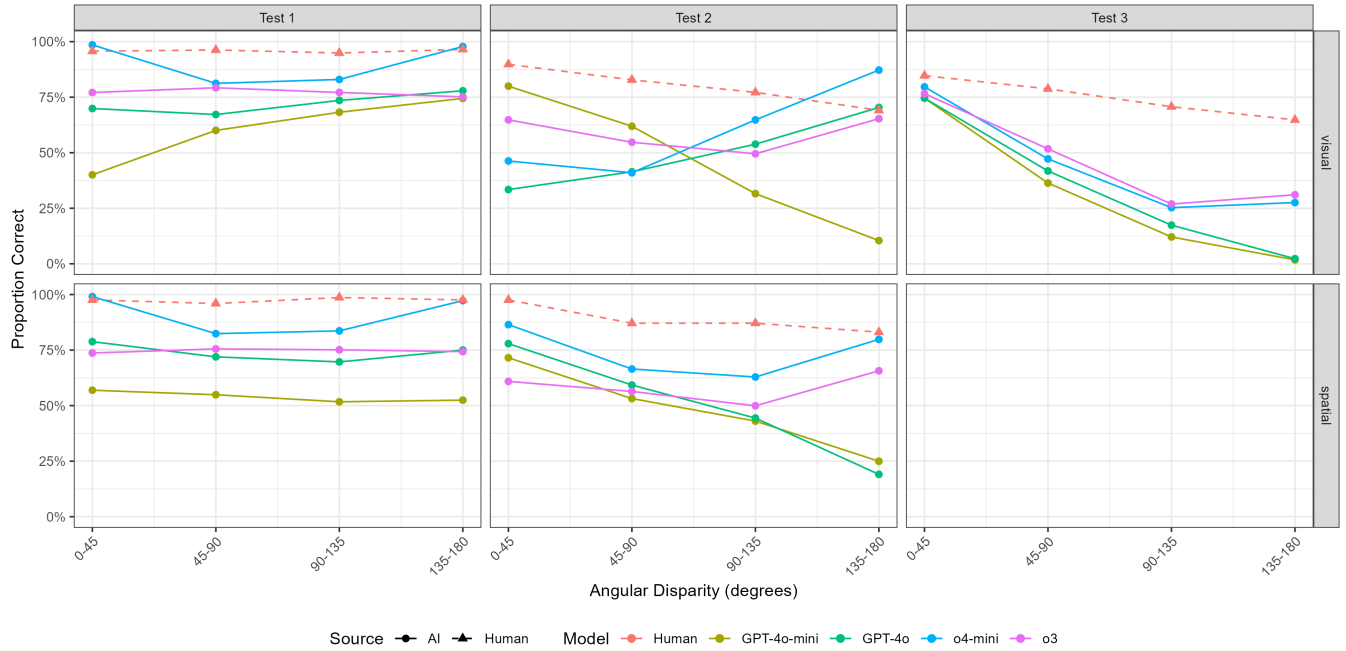


Figure 9: Mean accuracy for human participants (dashed, triangles) and MLMs (solid, circles) across the Rotating Figure Task test conditions, faceted by Test (rows) and question type (columns; Test 3 uses a single visuospatial question). The x -axis bins trials by angular disparity between viewer and figure. Humans stay at ceiling on Test 1 and show a graded decline on Tests 2–3, whereas models show non-human profiles.

References

- AI Security Institute, U. (2024, May). *Inspect AI: Framework for Large Language Model Evaluations*. https://github.com/UKGovernmentBEIS/inspect_ai
- Surtees, A., Apperly, I., & Samson, D. (2013). Similarities and differences in visual and spatial perspective-taking processes. *Cognition*, 129(2), 426–438.